

Ravi Kumar

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EDUCATION

Visvesvaraya Technological University (VTU)

Bachelor of Engineering in Information Science and Engineering

Belagavi, Karnataka

09/2023 – 07/2027

- **CGPA:** 8.1 / 10 | **Expected Graduation:** July 2027 (2027 batch)
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems (DBMS)

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), SQL

Backend: Spring Boot, Spring AI, Spring Security, Spring Data JPA, Hibernate, Spring Cloud Gateway, Eureka, Microservices, Event-Driven Architecture

Frontend: React, Vite, Tailwind CSS, React Router

APIs & Real-Time: RESTful APIs, JWT Authentication, WebSocket (STOMP), Server-Sent Events (SSE), Swagger/OpenAPI

Databases & Messaging: PostgreSQL, pgvector, MySQL, Redis, Apache Kafka

AI/ML Integration: Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, Spring AI, Azure OpenAI, Ollama

System Design: Distributed Systems, Event-Driven Sagas, Distributed Locking, Concurrency & Multithreading, Object-Oriented Design

DevOps & Cloud: Git, Maven, Docker, AWS (EC2), GitHub Actions (CI/CD)

Testing & Tools: JUnit, Postman

PROJECTS

Recall – Private Document Knowledge Workspace (RAG)

github.com/ravimehta251/Recall

Java 21 | Spring Boot 3 | Spring AI | PostgreSQL 16 + pgvector | React 19 | TypeScript | Azure OpenAI | JWT

- Built a full-stack RAG app where users upload PDF, DOCX, and TXT files and chat with an assistant that answers from retrieved chunks and cites its sources.
- Parsed and chunked uploaded files in memory with Spring AI, then embedded the chunks through Azure OpenAI (with local Ollama support) and stored them in PostgreSQL 16 using pgvector.
- Streamed chat replies token by token over authenticated SSE, sending token, citation, and completion events as the assistant responds.
- Secured the knowledge-space, document, and chat endpoints with JWT and Spring Security; created the frontend with React 19, TypeScript, Vite, Tailwind CSS, and Zustand.

Bidly – Real-Time Auction & Bidding Platform

github.com/ravimehta251/bidly

Java 21 | Spring Boot 3.3 | WebSocket (STOMP) | Redis (Redisson) | PostgreSQL | React 19

- Stopped double-selling on auctions by combining Redisson distributed locks with JPA optimistic locking, and confirmed it with a 200-thread concurrency test that produced zero conflicting writes.
- Broadcast live price and auction-close updates over WebSocket (STOMP), using Redis Pub/Sub so updates reach every backend instance behind an Nginx load balancer.
- Added REST endpoints for browsing, creating, and bidding on auctions behind a React UI, protecting the write operations with JWT and Spring Security.
- Tested the backend with JUnit 5, Spring Security Test, and Testcontainers, and served the production build through the same Nginx gateway as the API.

ShopMesh – Distributed Event-Driven E-Commerce Microservices Platform

github.com/ravimehta251/ShopMesh

Java 21 | Spring Boot 3.3 | Spring Cloud Gateway | Eureka | Kafka | MySQL | React 18 | Docker Compose | GitHub Actions

- Split checkout into 5 Spring Boot microservices (gateway, discovery, inventory, order, payment), routed through Spring Cloud Gateway with Eureka for service discovery, all running in Docker Compose.
- Modeled checkout as an event-driven saga using the transactional outbox pattern across 6 Kafka event types, with compensating events releasing reserved stock when payment fails.
- Separated data across 3 MySQL databases (inventory, order, payment) versioned with Flyway, and enforced stateless JWT authorization on every gateway-routed request.
- Set up a GitHub Actions workflow that deploys the full Docker Compose stack to an EC2 instance on every push to main.

ACHIEVEMENTS

- **2nd Runner-Up, JSSATE Hackathon:** Engineered Snap2Clean, a civic-issue reporting platform on React, Google Maps API, and MySQL, recognized for technical execution and social impact.
- **500+ DSA problems solved on LeetCode**, with sustained practice across arrays, strings, graphs, and trees.